INVERTER 11.1.1

FROM COMBINER BOXES	TO INVERTER	1-WAY DISTANCE	STRINGS	MODULE TYPE	FEEDER Isc AT STC	MAX. CURRENT PER NEC 690.8(A)(1)(2) [SITE-SPECIFIC]	MCA 690.8(B)	100% RATED DC DISCONNECT	OCPD 690.9(B)(1)	AL. DC FEEDER (90°C) UNDERGROUND (3) TYP.	AL. DC FEEDER (90°C) ABOVE GROUND (4) TYP.	Cu. EGC (2) TYP.	STUB UP 300.50(C) (1) TYP.	DC FEEDER AMPACITY	CB HARNESS AMPACITY	Vmp (STC)	Imp (STC)	OUTPUT CIRCUIT % VD	SOURCE CIRCUIT % VD	TOTAL % VD	INVERTER INPUT # (6)
11.1.1.C.1	INV 11.1.1	193'	96	FS-6470-P	250.6 A	273.6 A	342.0 A	275 A	350 A	600 kcmil	600 kcmil	#3 AWG.	2-1/2"	385 A	366 A	1146.6 V	236.2 A	0.28%	0.87%	1.15%	1
11.1.1.C.2	INV 11.1.1	281'	96	FS-6470-P	250.6 A	273.6 A	342.0 A	275 A	350 A	600 kcmil	600 kcmil	#3 AWG.	2-1/2"	385 A	366 A	1146.6 V	236.2 A	0.41%	0.87%	1.28%	2
11.1.1.C.3	INV 11.1.1	370'	80	FS-6470-P	208.8 A	228.0 A	285.0 A	250 A	315 A	500 kcmil	500 kcmil	#3 AWG.	2-1/2"	350 A	326 A	1146.6 V	196.8 A	0.54%	0.87%	1.41%	3
11.1.1.C.4	INV 11.1.1	443'	78	FS-6470-P	203.6 A	222.3 A	278.0 A	225 A	315 A	500 kcmil	500 kcmil	#3 AWG.	2-1/2"	350 A	326 A	1146.6 V	191.9 A	0.63%	0.87%	1.50%	4
TOTALS	–	–	350	FS-6470-P	913.5 A	997.5 A	1,247.0 A	–	–	–	–	–	–	–	–	–	AVERAGE	0.45%	0.87%	1.32%	–

INVERTER 11.1.2

FROM COMBINER BOXES	TO INVERTER	1-WAY DISTANCE	STRINGS	MODULE TYPE	FEEDER Isc AT STC	MAX. CURRENT PER NEC 690.8(A)(1)(2) [SITE-SPECIFIC]	MCA 690.8(B)	100% RATED DC DISCONNECT	OCPD 690.9(B)(1)	AL. DC FEEDER (90°C) UNDERGROUND	AL. DC FEEDER (90°C) ABOVE GROUND	Cu. EGC	STUB UP 300.50(C)	DC FEEDER AMPACITY	CB HARNESS AMPACITY	Vmp (STC)	Imp (STC)	OUTPUT CIRCUIT % VD	SOURCE CIRCUIT % VD	TOTAL % VD	INVERTER INPUT #
11.1.2.C.1	INV 11.1.2	183'	96	FS-6470-P	250.6 A	273.6 A	342.0 A	275 A	350 A	600 kcmil	600 kcmil	#3 AWG.	2-1/2"	385 A	366 A	1146.6 V	236.2 A	0.27%	0.87%	1.14%	1
11.1.2.C.2	INV 11.1.2	271'	96	FS-6470-P	250.6 A	273.6 A	342.0 A	275 A	350 A	600 kcmil	600 kcmil	#3 AWG.	2-1/2"	385 A	366 A	1146.6 V	236.2 A	0.39%	0.87%	1.26%	2
11.1.2.C.3	INV 11.1.2	360'	80	FS-6470-P	208.8 A	228.0 A	285.0 A	250 A	315 A	500 kcmil	500 kcmil	#3 AWG.	2-1/2"	350 A	326 A	1146.6 V	196.8 A	0.52%	0.87%	1.39%	3
11.1.2.C.4	INV 11.1.2	433'	80	FS-6470-P	208.8 A	228.0 A	285.0 A	250 A	315 A	500 kcmil	500 kcmil	#3 AWG.	2-1/2"	350 A	326 A	1146.6 V	196.8 A	0.63%	0.87%	1.50%	4
TOTALS	–	–	352	FS-6470-P	918.7 A	1,003.2 A	1,254.0 A	–	–	–	–	–	–	–	–	–	AVERAGE	0.44%	0.87%	1.31%	–

INVERTER 11.1.3

FROM COMBINER BOXES	TO INVERTER	1-WAY DISTANCE	STRINGS	MODULE TYPE	FEEDER Isc AT STC	MAX. CURRENT PER NEC 690.8(A)(1)(2) [SITE-SPECIFIC]	MCA 690.8(B)	100% RATED DC DISCONNECT	OCPD 690.9(B)(1)	AL. DC FEEDER (90°C) UNDERGROUND	AL. DC FEEDER (90°C) ABOVE GROUND	Cu. EGC	STUB UP 300.50(C)	DC FEEDER AMPACITY	CB HARNESS AMPACITY	Vmp (STC)	Imp (STC)	OUTPUT CIRCUIT % VD	SOURCE CIRCUIT % VD	TOTAL % VD	INVERTER INPUT #
11.1.3.C.1	INV 11.1.3	113'	94	FS-6470-P	245.3 A	267.9 A	335.0 A	275 A	350 A	600 kcmil	600 kcmil	#3 AWG.	2-1/2"	385 A	366 A	1146.6 V	231.2 A	0.16%	0.87%	1.03%	1
11.1.3.C.2	INV 11.1.3	114'	64	FS-6470-P	167.0 A	182.4 A	228.0 A	200 A	250 A	300 kcmil	300 kcmil	#4 AWG.	2"	260 A	234 A	1146.6 V	157.4 A	0.22%	0.87%	1.09%	2
11.1.3.C.3	INV 11.1.3	82'	101	FS-6470-P	263.6 A	287.9 A	360.0 A	300 A	400 A	600 kcmil	600 kcmil	#3 AWG.	2-1/2"	385 A	366 A	1146.6 V	248.5 A	0.12%	0.87%	0.99%	3
11.1.3.C.4	INV 11.1.3	83'	96	FS-6470-P	250.6 A	273.6 A	342.0 A	275 A	350 A	600 kcmil	600 kcmil	#3 AWG.	2-1/2"	385 A	366 A	1146.6 V	236.2 A	0.12%	0.87%	0.99%	4
TOTALS	–	–	355	FS-6470-P	926.6 A	1,011.8 A	1,265.0 A	–	–	–	–	–	–	–	–	–	AVERAGE	0.15%	0.87%	1.02%	–

INVERTER 11.1.4

FROM COMBINER BOXES	TO INVERTER	1-WAY DISTANCE	STRINGS	MODULE TYPE	FEEDER Isc AT STC	MAX. CURRENT PER NEC 690.8(A)(1)(2) [SITE-SPECIFIC]	MCA 690.8(B)	100% RATED DC DISCONNECT	OCPD 690.9(B)(1)	AL. DC FEEDER (90°C) UNDERGROUND	AL. DC FEEDER (90°C) ABOVE GROUND	Cu. EGC	STUB UP 300.50(C)	DC FEEDER AMPACITY	CB HARNESS AMPACITY	Vmp (STC)	Imp (STC)	OUTPUT CIRCUIT % VD	SOURCE CIRCUIT % VD	TOTAL % VD	INVERTER INPUT #
11.1.4.C.1	INV 11.1.4	187'	96	FS-6470-P	250.6 A	273.6 A	342.0 A	275 A	350 A	600 kcmil	600 kcmil	#3 AWG.	2-1/2"	385 A	366 A	1146.6 V	236.2 A	0.27%	0.87%	1.14%	1
11.1.4.C.2	INV 11.1.4	275'	96	FS-6470-P	250.6 A	273.6 A	342.0 A	275 A	350 A	600 kcmil	600 kcmil	#3 AWG.	2-1/2"	385 A	366 A	1146.6 V	236.2 A	0.40%	0.87%	1.27%	2
11.1.4.C.3	INV 11.1.4	363'	96	FS-6470-P	250.6 A	273.6 A	342.0 A	275 A	350 A	600 kcmil	600 kcmil	#3 AWG.	2-1/2"	385 A	366 A	1146.6 V	236.2 A	0.53%	0.87%	1.40%	3
11.1.4.C.4	INV 11.1.4	451'	78	FS-6470-P	203.6 A	222.3 A	278.0 A	225 A	315 A	500 kcmil	500 kcmil	#3 AWG.	2-1/2"	350 A	326 A	1146.6 V	191.9 A	0.64%	0.87%	1.51%	4
TOTALS	–	–	366	FS-6470-P	955.3 A	1,043.1 A	1,304.0 A	–	–	–	–	–	–	–	–	–	AVERAGE	0.45%	0.87%	1.32%	–

INVERTER 11.1.5

FROM COMBINER BOXES	TO INVERTER	1-WAY DISTANCE	STRINGS	MODULE TYPE	FEEDER Isc AT STC	MAX. CURRENT PER NEC 690.8(A)(1)(2) [SITE-SPECIFIC]	MCA 690.8(B)	100% RATED DC DISCONNECT	OCPD 690.9(B)(1)	AL. DC FEEDER (90°C) UNDERGROUND	AL. DC FEEDER (90°C) ABOVE GROUND	Cu. EGC	STUB UP 300.50(C)	DC FEEDER AMPACITY	CB HARNESS AMPACITY	Vmp (STC)	Imp (STC)	OUTPUT CIRCUIT % VD	SOURCE CIRCUIT % VD	TOTAL % VD	INVERTER INPUT #
11.1.5.C.1	INV 11.1.5	189'	96	FS-6470-P	250.6 A	273.6 A	342.0 A	275 A	350 A	600 kcmil	600 kcmil	#3 AWG.	2-1/2"	385 A	366 A	1146.6 V	236.2 A	0.28%	0.87%	1.15%	1
11.1.5.C.2	INV 11.1.5	278'	96	FS-6470-P	250.6 A	273.6 A	342.0 A	275 A	350 A	600 kcmil	600 kcmil	#3 AWG.	2-1/2"	385 A	366 A	1146.6 V	236.2 A	0.40%	0.87%	1.27%	2
11.1.5.C.3	INV 11.1.5	366'	80	FS-6470-P	208.8 A	228.0 A	285.0 A	250 A	315 A	500 kcmil	500 kcmil	#3 AWG.	2-1/2"	350 A	326 A	1146.6 V	196.8 A	0.53%	0.87%	1.40%	3
11.1.5.C.4	INV 11.1.5	440'	78	FS-6470-P	203.6 A	222.3 A	278.0 A	225 A	315 A	500 kcmil	500 kcmil	#3 AWG.	2-1/2"	350 A	326 A	1146.6 V	191.9 A	0.62%	0.87%	1.49%	4
TOTALS	–	–	350	FS-6470-P	913.5 A	997.5 A	1,247.0 A	–	–	–	–	–	–	–	–	–	AVERAGE	0.45%	0.87%	1.32%	–

GENERAL NOTES

1. ALL EQUIPMENT SHALL BE UL LISTED.
2. MULTIPLE CONDUCTORS SHALL BE PERMITTED FOR USE IN ANY OF THE WIRING METHOD. DERATING FACTORS SHALL APPLY TO MORE THAN THREE (3) CURRENT-CARRYING CONDUCTORS IN RACEWAY OR CABLE. THE ALLOWABLE CAPACITY OF EACH CONDUCTOR SHALL BE REDUCED AS SHOWN IN NEC TABLE 310.15(B)(3)(a). NUMBER OF CONDUCTORS INSTALLED IN CONDUIT SHALL NOT EXCEED PERMITTED PERCENTAGE FILL SPECIFIED IN NC TABLE 1, CHAPTER 9.
3. THE AVERAGE VOLTAGE DROP FOR DC CIRCUITS FROM MODULE TO INVERTER SHALL NOT EXCEED 1.5%. CONTRACTOR SHALL NOTIFY ENGINEER IF ACTUAL ROUTING EXCEEDS THE DESIGNED ONE-WAY DISTANCES SHOWN IN DC WIRE SCHEDULE AS SIZE OF CONDUCTORS AND CONDUITS MAY HAVE TO BE INCREASED. VOLTAGE DROP EVALUATED AS A POWER LOSS AT STC CONDITIONS. Imp CURRENT IS USED TO DETERMINE $I^2 \times R$ LOSSES ON EVERY CONDUCTOR SECTION FROM SOURCE CIRCUIT OUTPUT TO INVERTER INPUT. TOTAL LOSS DIVIDED BY TOTAL STC NAMEPLATE IS THE LOSS PERCENTAGE.
4. CONTRACTOR SHALL TORQUE ALL ELECTRICAL CONNECTIONS ACCORDING TO MANUFACTURER'S INSTALLATION INSTRUCTION.
5. CABLE SIZING MAY ACCOMMODATE MODULE CLASS UP TO 480W OF FS6 MODULES

KEYED NOTES

1. DIRECT BURIAL INSTALLATION. CONDUIT OR GREENTROUGH TO BE USED FOR WIRING PROTECTION WHILE EMERGING FROM THE GROUND. FOR MULTIPLE FEEDERS IN A CONDUIT REFER TO PV WIRE CONDUIT FILL TABLE ON THIS SHEET. FOR ABOVE/BELOW GROUND TRANSITIONS REFER TO GREENTROUGH TRANSITION FILL TABLE ON THIS SHEET
2. #2 CU. COMMON GND. SHALL BE LOCATED IN EACH DC TRENCH AND RUN TO EACH COMBINER BOX.
3. UNDERGROUND DC FEEDER AMPACITY FROM NEC TABLE 310.15(B)(16). NO AMPACITY ADJUSTMENTS ARE REQUIRED FOR CONDITIONS OF USE.
4. ABOVE GROUND COMBINER HARNESS AMPACITY DETERMINED USING NEC TABLE 310.15(B)(20) FOR MESSENGER SUPPORTED WIRING UP TO 40°C. PER NEC 310.15(B)(3)(A), A 0.7 AMPACITY ADJUSTMENT IS APPLIED TO ACCOMMODATE BUNDLING OF UP TO (6) CONDUCTORS WITHIN THE CABLE HANGERS. PER NEC 310.15(B)(2)(b) A 0.89 AMPACITY ADJUSTMENT IS APPLIED TO ACCOMMODATE A 50°C AMBIENT TEMPERATURE.

PV WIRE CONDUIT FILL

MAXIMUM NUMBER OF 2kV PV WIRE CONDUCTORS PER CONDUIT

	2"	3"	4"	5"
250 kcmil	2	6	6	6
300 kcmil	–	4	6	6
350 kcmil	–	4	6	6
400 kcmil	–	2	4	6
500 kcmil	–	2	5	6
600 kcmil	–	2	4	6
750 kcmil	–	2	4	6

NOTE: TABLE REPRESENTS MAX 40% FILL AND INCLUDES (1) #2 USE-2 EQUIPMENT GROUNDING CONDUCTOR PER CONDUIT. NO MORE THAN (6) DC CABLES ARE PERMITTED IN A CONDUIT WITHOUT WRITTEN ENGINEERING APPROVAL. THIS TABLE IS NOT APPLICABLE WHEN QUANTITIES ARE SPECIFIED ON BORING OR TRENCHING DETAILS ON E-9.X SERIES DRAWINGS.

GREENTROUGH TRANSISTION FILL

MAXIMUM NUMBER OF CONDUCTORS PER TRANSITION

SIZE	150B	200B	300B
250 kcmil	16	24	24
300 kcmil	14	24	24
350 kcmil	14	24	24
400 kcmil	10	18	24
500 kcmil	10	18	24
600 kcmil	8	14	20
750 kcmil	6	12	16

NOTE: PER NEC 310.15(A)(2) CONDUCTORS ARE NOT SUBJECT TO AMPACITY ADJUSTMENT THROUGH THE GREENTROUGH TRANSITION WHEN TOTAL CONDUCTOR LENGTH IS GREATER THAN 33' (LESS THAN 10% OF TOTAL CABLE LENGTH). IN ADDITION TO DC CABLES, A #2 USE-2 EGC SHALL BE ROUTED THROUGH EACH TRANSITION.

COMBINER BOXES NAMING CONVENTION

X . Y . Z . C . # (EXAMPLE: 1.2.1.C.6)

- # — COMBINER BOXES NUMBER
- C — EQUIPMENT DESIGNATION
- Z — INVERTER NUMBER
- Y — BLOCK NUMBER
- X — MV CIRCUIT NUMBER

IFC

SOLV ENERGY

CONSULTANT

BLYMYER ENGINEERS
1101 MARINA VILLAGE PARKWAY #100
ALAMEDA, CA 94501 510.521.2773

STAMP/SEAL

REGISTERED PROFESSIONAL ENGINEER JOHN A. HEIN No. E20055 ELECTRICAL STATE OF CALIFORNIA

REV #	DESCRIPTION	DATE
0	ISSUED FOR PERMIT	04/08/2022
1	ISSUED FOR CONSTRUCTION	06/23/2022
2	CONSTRUCTION CHANGES	08/26/2022
3	CONSTRUCTION UPDATES	11/30/2022
4	CONSTRUCTION UPDATES	12/03/2022
5	OWNER COMMENTS	02/07/2023
6	CONSTRUCTION UPDATES	11/13/2023

RECORD DRAWING
BLYMYER ENGINEERS, INC.
11/20/2023

THESE RECORD DRAWINGS (AS-BUILTS) ARE BASED UPON MARKED-UP PRINTS, DRAWINGS, AND OTHER DATA FURNISED TO THE ENGINEER BY SOLV ENERGY AND INTERSECT POWER. THE DESIGN PROFESSIONAL HAS NOT VERIFIED THE ACCURACY AND/OR COMPLETENESS OF THIS INFORMATION AND SHALL NOT BE RESPONSIBLE FOR ANY ERRORS OR OMISSIONS WHICH MAY BE INCORPORATED HERIN AS A RESULT.

PROJECT TITLE:
OBERON I & II
SINGLE-AXIS TRACKER
PHOTOVOLTAIC ARRAYS
500 MWAC
RIVERSIDE COUNTY, CA

SHEET TITLE:
BLOCK 11.1
DC WIRE & CONDUIT SCHEDULE

JOB NO.: 222011 | PROJECT MGR: SG
DRAWN: SGC | SCALE: NTS
SHEET NUMBER
E-4.11.1

Z:\2022\222011 SOLV Oberon 1-2 CA\01 CAD\01 PV\As-Built\Rev 6\working - PVLE-4.11.y.dwg November 19, 2023 - 5:08PM VCHLBAR